

Phase 1 Guide (KT1 — First Checkpoint): Movie NLP Success Analyzer

Course: Systems for Research and Data Analysis (SIAP)

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Checkpoint: KT1 (First checkpoint)

This document describes **what KT1 requires** and **how to deliver** the Movie NLP Success Analyzer: checklist, code locations, rationale, and links.

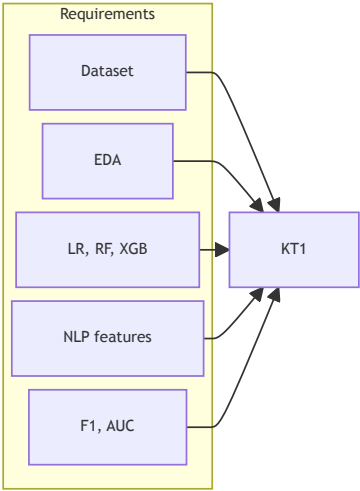
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1. KT1 Requirements

(SIAP — Systems for Research and Data Analysis.)

Requirement	Why it matters for this project
Complete dataset	TMDB metadata + OpenSubtitles text + Academy Awards labels; merged and cleaned for modeling.
EDA (Exploratory Data Analysis)	Understand distributions, correlations, class balance, and subtitle coverage before modeling. EDA overview
Baseline models	At least LR (Logistic Regression) , RF (Random Forest) , XGBoost — fixed hyperparameters; no tuning required at KT1. sklearn
NLP features	Extract features from subtitle text: TF-IDF (e.g. SVD), sentiment (e.g. VADER), lexical diversity; combine with metadata.
Class imbalance handling	SMOTE and/or class_weight for award prediction (minority class ~13%). SMOTE
Evaluation	Stratified 70/15/15 split; F1 , AUC (and optionally balanced accuracy); no optimization required.
Presentation	~10 min: intro → data → EDA → NLP features → baseline models → conclusions.



2. KT1 Deliverables — Checklist

Item	Where	Why
Dataset	data/processed/ : final_movie_list.csv , TMDB + subtitles + awards merged	Single modeling table with targets and IDs for NLP join.
EDA	01_eda.ipynb	Target distributions, budget/revenue, genre, temporal trends, subtitle coverage, correlations.
NLP features	02_nlp_features.ipynb + src/nlp_features.py	Text stats, sentiment, lexical diversity, TF-IDF matrix; saved for 03 .
Baseline models	03_baseline_models.ipynb + src/models.py	LR, RF, XGB on progressive feature sets; SMOTE/class_weight for award.
Feature sets	1_text_basic → 3_text_full (NLP + SVD) → 4_+genre ... 8_+vote_revenue	Revenue excluded from commercial sets (data leakage).
Metrics	F1, F1@opt, AUC, balanced accuracy; tables and heatmaps	Reproducible evaluation; F1@opt for imbalanced award task.
Split	70% train / 15% val / 15% test, stratified	Required by course; val used for threshold selection.
Report	phase_01/README.md	Problem, data, EDA, NLP, baseline results, conclusions.

3. Recommended Work Order (KT1)

3.1 Phase A — Data & EDA

1. Dataset assembly

Ensure `data/processed/` has merged TMDb + awards; add subtitle coverage and `is_commercial / is_award_winner` labels. Fix **data leakage**: `is_commercial = (revenue/budget ≥ 2)` → do **not** use revenue in commercial feature sets.

Why: Clean targets and leakage-free features are mandatory before any model.

2. EDA notebook (`01_eda.ipynb`)

Overview, target distributions, financials, genre, temporal trends, subtitle coverage, correlation analysis (e.g. point-biserial with targets).

Why: Shows you understand the data and supports methodology in the report.

3. NLP feature extraction (`02_nlp_features.ipynb` , `src/nlp_features.py`)

Load subtitle texts, clean, compute: word/sentence stats, lexical diversity, sentiment (e.g. VADER), TF-IDF; save `nlp_features.csv` and TF-IDF matrix (+ IDs) for modeling.

Why: Baseline models in 03 need these features; TF-IDF + SVD used in set 3_text_full.

3.2 Phase B — Baseline Models

4. Data loading & feature sets (`03_baseline_models.ipynb`)

Load main dataset + NLP features, merge, create genre dummies, filter to valid budget/revenue/runtime + subtitle. Define **eight progressive feature sets** (`1_text_basic ... 8_+vote_revenue`); keep commercial sets free of revenue.

Why: Isolates contribution of each factor (text vs metadata).

5. Train & evaluate

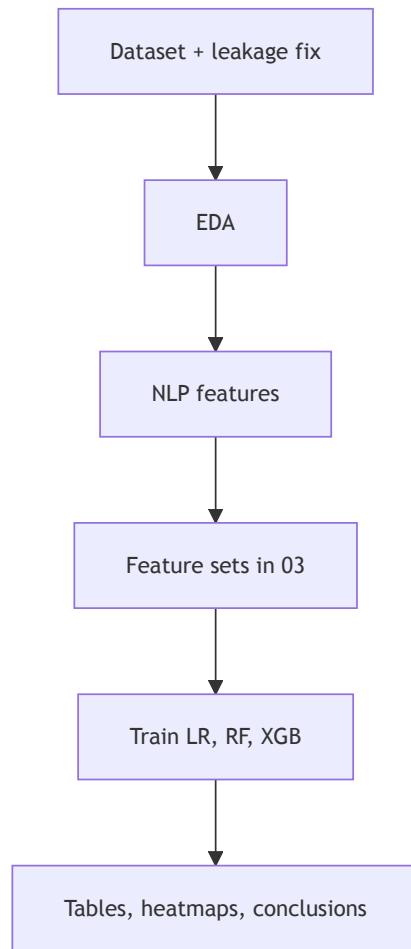
Use `create_splits` , `get_baseline_models` , `train_baseline` from `src/models.py` . For each set and both targets: stratified split, optional SMOTE for award, train LR/RF/XGB; collect F1, F1@opt, AUC, balanced accuracy.

Why: Single pipeline; results in one table and heatmaps.

6. Conclusions

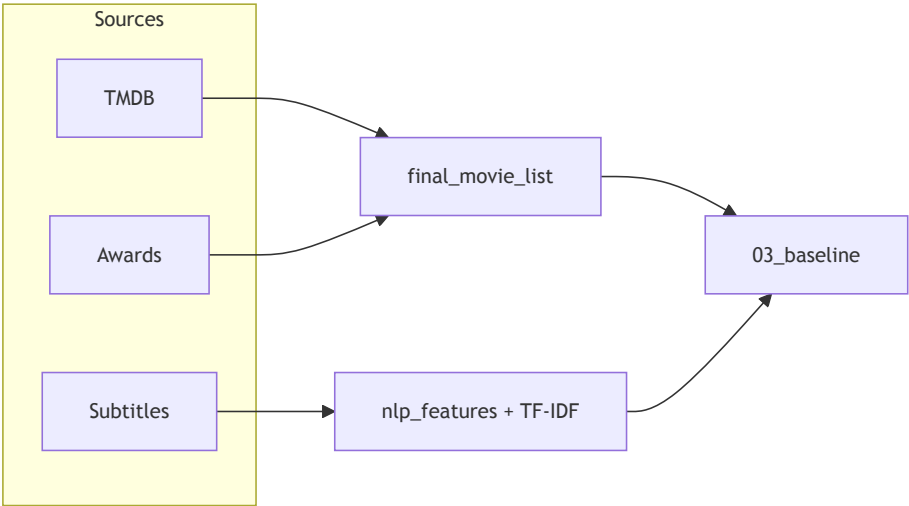
Short summary: which feature set gives best F1/AUC; commercial vs award; note that parameters are not optimized (KT2).

Why: Directly answers “what did KT1 achieve?” for the presentation and report.



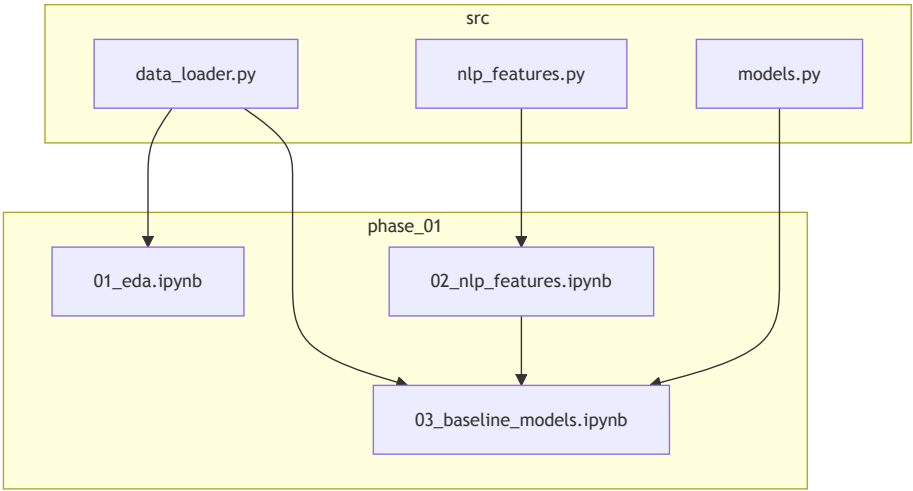
4. Data & NLP Pipeline (KT1 Scope)

Step	Source	Output	Used in
TMDB metadata	API / data/raw/	Budget, revenue, runtime, genres, vote_count, release_date	EDA, modeling
Awards	CSV	is_award_winner	Target
Commercial label	Derived	is_commercial = (revenue/budget ≥ 2)	Target; exclude revenue from commercial features
Subtitles	OpenSubtitles / .txt	Raw text per movie	02_nlp_features
NLP features	nlp_features.py	word_count, sentiment, lexical, TF-IDF matrix	03_baseline_models
SVD	03_baseline_models.ipynb	N_SVD components (e.g. 100) from TF-IDF	Set 3_text_full



5. Code Locations

Goal	File	Action
Load & merge data	src/data_loader.py	load_main_dataset , load_nlp_features , merge_datasets
Genre dummies	src/data_loader.py	get_unique_genres , create_genre_dummies
EDA	01_eda.ipynb	Full EDA; figures in figures/
NLP extraction	src/nlp_features.py	Text cleaning, sentiment, lexical, TF-IDF; batch processing
NLP notebook	02_nlp_features.ipynb	Build nlp_features.csv, tfidf_matrix.npz, corpus IDs
Splits & models	src/models.py	prepare_modeling_data , create_splits , get_baseline_models , train_baseline
Results & plots	src/models.py	results_table , f1_at_optimal_threshold , plot_confusion_matrices , plot_roc_curves
Baseline pipeline	03_baseline_models.ipynb	Define FEATURE_SET_CONFIG, run train/eval, heatmaps, SMOTE vs class_weight



6. Evaluation Targets (KT1 Baseline)

Task	KT1 expectation	Note
Commercial	F1 ~0.60–0.68, AUC ~0.70–0.78 (set 7_+runtime)	Revenue excluded; budget, runtime, vote_count, genre, NLP.
Award	F1(opt) ~0.50–0.55, AUC ~0.78–0.87	Imbalanced; use F1 at optimal threshold; SMOTE or class_weight.

Results **do not need to be strong** at KT1; the goal is a **reproducible baseline** and clear methodology.
Hyperparameter tuning and SVM are for KT2.

7. Presentation & Report

- **Presentation:** ~10 min. Order: intro → data → EDA → NLP features → baseline models → conclusions. See [Phase_01_KT1_Defense_SR.md](#).
- **Report:** README.md in phase_01 — problem, data, EDA, NLP, baseline results, conclusions.
- **Code:** This repo; deliverable for KT1 is phase_01/ (notebooks 01–03 + src/).

This document is the **Phase 1 implementation guide** and supports consistent delivery for KT1.